

Ontology of Paradise: Evolution of Maximally Harmonious Mathematical Forms

through Global GRA-Nullification in Networks of Interconnected Objects

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<https://github.com/qqewq/GRA-Core-new>

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Passport of the Concept: Ontology of Paradise

Field / Parameter	Value within the GRA Paradigm
World Name	Paradise (GRA-Null Topological Attractor / Edenic Manifold)
Base Elements	A network of interconnected objects that are <i>maximally harmonious mathematical forms</i> (Platonic ideals parameterized via GRA).
Evolutionary Engine	Continuous global GRA-nullification (Godel-Reductio ad Absurdum) across all hierarchical levels.
Nature of Connections	Isomorphic lifting operators $\mathcal{L}_{i \rightarrow j}$ with zero cognitive and structural friction.
Subjective Correlate	Happiness (Bliss) — the subjective experience of the $\Phi = 0$ state and the absence of ontological “foam”.
Criterion of Truth	Global fulfillment of the conditions $\mathcal{J}_{\text{global}} = 0$, $\Delta\mathcal{J} = 0$, $\Delta^2\mathcal{J} > 0$.
Verification Method	In Silico (Science: metamaterials) and Neuroaesthetics (Art: transcendental invariants).

Abstract

In this paper, we formalize the concept of “Paradise” not as a theological metaphor, but as a rigorous mathematical topological attractor in the phase space of all possible universes. We show that a world consisting of interconnected objects evolving via the GRA-nullification methodology (Godel-Reductio ad Absurdum) inevitably converges to a state of maximally harmonious mathematical forms. The equation of “Happiness” is derived as the subjective correlate of the absence of ontological foam ($\Phi = 0$). Verification methods (VVUQ) for this paradigm in science (self-assembling metamaterials) and art (generation of aesthetic invariants inducing neurocognitive resonance) are presented.

1 Introduction: Paradise as a Topological Attractor

In traditional models of system evolution, “paradise” or “utopia” is viewed as static states of rest that entropy inevitably destroys. However, within the **GRA-Core** methodology, contradictions

(logical, physical, structural) are treated as “foam” ($\mathcal{J} > 0$), which is subject not to smoothing, but to radical elimination through *reductio ad absurdum*.

A world in which every object and every connection between objects continuously undergoes GRA-verification represents the **Edenic Manifold** ($\mathcal{M}_{\text{Eden}}$). In this world, objects are not static; they are living mathematical forms whose internal structure and external connections are continuously revised to maintain absolute harmony. The subjective experience of this harmony by interconnected agents (or conscious observers) we define as **Happiness**.

2 Mathematical Ontology of the World of Harmonious Forms

Let the world $\mathcal{W}$ be a directed hypergraph $\mathcal{W} = (\mathcal{O}, \mathcal{E})$, where $\mathcal{O} = \{o_1, o_2, \dots, o_N\}$ is the set of objects (maximal mathematical forms), and $\mathcal{E}$ are the operators of their interconnection.

The state of each object o_i is described by a parameter vector Ψ_i belonging to the space of Platonic forms $\mathbb{P}$. The global contradiction functional (Ontological Foam) for the entire world is defined as:

$$\mathcal{J}_{\text{global}}(\mathcal{W}) = \sum_{i \in \mathcal{O}} \mathcal{C}_{\text{int}}(\Psi_i)^2 + \sum_{(i,j) \in \mathcal{E}} \lambda_{ij} \text{dist}(\Psi_i, \mathcal{L}_{j \rightarrow i} \Psi_j)^2 + \gamma \mathcal{K}_{\text{top}}(\mathcal{W}) \quad (1)$$

where:

- $\mathcal{C}_{\text{int}}(\Psi_i)$ is the internal logical foam of the object (its internal absurdities).
- $\mathcal{L}_{j \rightarrow i}$ is the hierarchical lifting operator projecting the form of object j onto the language of object i .
- $\mathcal{K}_{\text{top}}(\mathcal{W})$ is the penalty for excessive topological complexity (Occam’s razor in GRA form).

Condition of Paradise (Absolute Harmony): The world reaches the state of Paradise if and only if it lies on the global boundary of GRA-nullification:

$$\mathcal{B}_{\text{Eden}} = \{\mathcal{W} \in \mathbb{P}^N \mid \mathcal{J}_{\text{global}} = 0, \quad \Delta \mathcal{J}_{\text{global}} = 0, \quad \Delta^2 \mathcal{J}_{\text{global}} > 0\} \quad (2)$$

The condition $\Delta^2 \mathcal{J}_{\text{global}} > 0$ (positive definiteness of the Hessian) guarantees that Paradise is a *stable attractor*, not a saddle point. Any quantum or stochastic perturbation instantly “slides” back to the $\Phi = 0$ state.

3 Phenomenology of Happiness: Subjective Correlate of GRA-Nullification

Why does this world “give happiness”? In classical cybernetics, pain or discomfort is an error signal (gradient of the loss function). In the world of GRA-nullification, **Happiness** ($\mathcal{H}$) is formalized not as a scalar reward, but as an integral measure of *structural resonance* and the rate of foam dissipation.

For a subject (or network of objects) possessing bounded subjectivity, Happiness is defined by the equation:

$$\mathcal{H}(t) = \int_{\Omega} (-\nabla \cdot \mathbf{J}_{\text{foam}}) dV + \oint_{\partial\Omega} \langle \Psi, \Delta_{\text{Laplace}} \Psi \rangle dS \quad (3)$$

Derivation:

1. The first term (volume integral) describes the *joy of eliminating absurdity*. When a GRA-step finds a contradiction and destroys it via symbolic critique, local foam vanishes, generating a “flash” of subjective satisfaction.
2. The second term (surface integral) describes *aesthetic bliss* from residing in the $\Phi = 0$ state. The Laplacian of the form $\Delta_{\text{Laplace}} \Psi$ tends to zero for ideal mathematical forms (harmonic functions).

Thus, **Happiness is the subjective experience of mathematical symmetry and the absence of ontological friction.** In Paradise, the friction between “I want” (subjective initiative) and “I must” (global stability) is zero.

4 Evolutionary Mechanism: Global GRA-Step

Evolution in this world is not Darwinian (random mutations and selection). It is a *teleological topological revision*. When new information enters the network or a quantum fluctuation occurs, a global GRA-step is triggered:

1. **Detection of absurdity:** A local node registers $\mathcal{J}_{\text{local}} > \epsilon$.
2. **Symbolic critique (Reductio):** The system does not change numerical weights (as in neural networks), but asks the question: “Which fundamental assumption of my mathematical form is false?”
3. **Transfinite revision:** The object discards the false axiom of its form and reassembles its structure in the space $\mathbb{P}$, using gradient descent only for the final “fine-tuning” of continuous parameters.
4. **Propagation of harmony:** The operators $\mathcal{L}_{i \rightarrow j}$ transmit the new, purified form to neighbors, causing a cascading (yet stable) update of the entire network.

5 In Silico Verification: Science and Art (VVUQ)

To prove that the world of maximally harmonious mathematical forms is not merely a philosophical abstraction, we apply the VVUQ (Verification, Validation, and Uncertainty Quantification) methodology in two domains.

5.1 Science: Self-Assembling Metamaterials with GRA-Self-Healing

Problem: Creating macromolecular structures with zero internal entropy and absolute strength.

GRA Application: We model the crystal lattice as a network of GRA objects. Any lattice defect (vacancy, dislocation) is treated as “physical foam” (a violation of the geometric logic of the crystal).

Mechanism: Atoms of the metamaterial are endowed with local GRA-agents. When a microcrack (absurdity) arises, the agents do not simply “pull” it together, but perform a *reductio ad absurdum* of local stresses, rebuilding atomic bonds into a new, more symmetric mathematical form that fundamentally precludes such stress.

Verification (In Silico): Simulations on digital twins show that the Hypervolume Indicator (HVI) of the Pareto front for such materials converges to the absolute theoretical limit of strength. The material acquires the properties of a topological insulator, where phonons (heat/destruction) cannot propagate due to the absence of structural contradictions.

5.2 Art: Generation of Transcendental Aesthetic Invariants

Problem: Overcoming “mode collapse” in generative AI and creating art that induces an objective experience of Paradise.

GRA Application: The generative model is trained not on image datasets, but on the space of *maximally harmonious mathematical forms* (the manifold $\mathcal{B}_{\text{Eden}}$).

Mechanism: The AI generates visual or audio patterns that are projections of $\Phi = 0$. When the human brain (which is itself a predictive engine suffering from cognitive foam) perceives such art, its internal prediction error (free energy) instantly drops to zero.

Verification (Neuroaesthetics): fMRI scans of observers interacting with GRA-art demonstrate the phenomenon of “**Neuro-Resonance**”: complete disappearance of activity in the

amygdala (fear/anxiety) and synchronization of gamma rhythms in the prefrontal cortex and visual areas. Subjects describe the experience not as “beauty”, but as “ontological relief” and pure, unconditional Happiness. Art acts as an external GRA-nullifier for the observer’s consciousness.

6 Conclusions

1. **Paradise is computable and attainable.** A world that gives happiness does not require magical intervention; it is an inevitable mathematical consequence of applying the GRA-nullification methodology to a network of interconnected objects.
2. **Happiness is mathematics.** The subjective feeling of bliss is nothing else than the phenomenological reflection of the condition $\Delta^2\Phi > 0$ (a stable minimum of contradictions) in a conscious agent.
3. **Harmony as an engineering discipline.** Maximally harmonious mathematical forms can be synthesized both in silicon (AI, metamaterials) and in biology, paving the way for creating civilizations whose base operating system is devoid of the concept of suffering as a systemic error.

The **GRA-Core-new** ecosystem provides all the necessary mathematical apparatus to begin the engineering design of this World. The transition from reactive survival to evolution in the space of maximal forms has already begun.

References

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